Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Research engineer and computational scientist with a PhD in Physics and over ten years of experience building physics-based numerical models (PDE solvers, radiative transfer, MHD) and applying machine learning to physical-science problems. Python-fluent, with GPU and HPC experience and an open-source track record: architected and maintained simulation libraries adopted by other research groups, tested and shipped code through CI/CD with test suites, and mentored students. Strong at pairing physical principles with data-driven models and explaining the results to scientific and engineering audiences.

Skills

Languages: Python, C/C++, Fortran 95, Rust, R, Java, SQL, Julia, Shell

ML & Data: Scientific machine learning, hybrid physics–ML modeling, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, neural networks, random forest, SVM

Modeling: Numerical PDEs (advection–diffusion, Fokker–Planck), finite difference/volume methods, multiphysics simulation, uncertainty quantification

HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5

Cloud & Tools: Git and code review, Docker, Azure, CI/CD (Jenkins), Kafka, Jira, QGIS, ParaView, VisIt

Experience

Independent Contractor **Jan 2026 – Present**
Mercor Remote
Physics subject-matter expert evaluating frontier language models for a leading AI lab.
- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.

Spatial Data Scientist **May 2025 – Jan 2026**
TealWaters Seattle, WA
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Independent Researcher & Open-Source Developer **Jan 2024 – May 2025**
Self-directed Seattle, WA
Continued simulation research and open-source development between industry roles.
- Co-developed Tleco, an open-source Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer for relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library for wind-driven spore dispersal over hilly forest terrain (RcppParallel, OpenMP).

Software Engineer **Apr 2022 – Jan 2024**
Paychex Remote
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.
- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Postdoctoral Research Associate **Feb 2021 – Apr 2022**
Rochester Institute of Technology Rochester, NY
HPC simulations of supermassive black hole binaries in an NSF-funded project.
- Led a 7-person team in an NSF-sponsored upgrade of PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries, adding black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, identifying performance bottlenecks in the parallel code to make it run efficiently on HPC clusters.
- Calibrated Paramo parameters against Fermi-LAT observations using gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow **Oct 2018 – Nov 2020**
Purdue University West Lafayette, IN
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer. OpenMP gives about 20× speedup on 5 cores; GPU offload tested with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; released and documented Paramo as open source, and other groups have adopted it, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals, one funded ($68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow **Jan 2018 – Sep 2018**
Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Graduate Research Assistant **Oct 2011 – Jul 2017**
Universitat de València Valencia, Spain

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Projects

Blazar Population ML **Jan 2026 – Present**
Physics-informed ML (Python, scikit-learn, PyTorch) github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Training a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, validated against Tleco output.
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

Selected Publications (9 peer-reviewed articles, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

Education

Universitat de València **Valencia, Spain**
Ph.D. in Physics, Excellent Cum Laude 2017

Universidad Michoacana de San Nicolás de Hidalgo **Morelia, Mexico**
M.Sc. in Physics

Universidad Autónoma del Estado de México **Toluca, Mexico**
B.Sc. in Physics